

# Kyle M. Booker

Vancouver, B.C., Canada

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## Experience

### Reviewer, AI Training (Programming, Mathematics)

Vancouver, B.C.

Outlier AI

May 2024 - Present

- Ran, debugged, and fixed contributor-submitted Python, C, C++, and JavaScript code across math, programming, and physics tasks to verify correctness, edge-case handling, and runtime behaviour before submissions enter the training pipeline.
- Audited 5,000+ submissions for input and output quality, identifying logical errors and bugs and authoring structured feedback that improved contributor consistency and downstream model accuracy.

### Data Scientist

Vancouver, B.C.

IntelliFi - Residential Technology

Jun 2022 - Mar 2023

- Delivered a rent-roll OCR ingestion pipeline on AWS using S3, Lambda, Textract, and EC2 processing 1,000+ monthly PDFs end-to-end, replacing manual data entry and cutting analyst workload by 70%.
- Built and deployed supervised models on AWS forecasting probability of delinquency for residential loans, with REST inference endpoints integrated into production systems.
- Owned the reporting layer over the data pipeline, exposing KPIs to product, sales, and executive stakeholders via Power BI.

### Data Analyst

Vancouver, B.C.

CMLS Financial - Residential Capital Markets

Jun 2021 - Jun 2022

- Built and operated an Airflow-orchestrated Python/SQL ETL pipeline processing 1 GB of customer data per hour into PostgreSQL, replacing manual workflows with scheduled, monitored DAGs.
- Developed dbt models transforming raw data into analytics-ready tables that powered Power BI reporting for business stakeholders.
- Engineered a Selenium ingestion service (Python) scraping 30,000 real-estate listings per week into a PostgreSQL store for analytics.
- Delivered a production churn-prediction service (Python, pandas) at 80% recall, driving a 15% lift in retention via targeted campaigns.

### Graduate Research Assistant

Waterloo, Ont.

University of Waterloo

Sep 2018 - Apr 2021

- Implemented H(div)-conforming discontinuous Galerkin methods for multiphase flow in C++ on the MFEM finite-element library: 2,000+ lines including custom discretizations, solvers, and numerical experiments.
- Ran large-scale simulations on Compute Canada HPC clusters with parallel jobs spanning many cores and multi-day runtimes; built Python and ParaView pipelines for post-processing, convergence analysis, and visualization.

## Skills

**Programming** Python, SQL, Bash, JavaScript, TypeScript, C, C++, Java

**Data Engineering** Apache Airflow, dbt, ETL/ELT pipelines, PostgreSQL, pandas, MongoDB, Redis, Power BI

**Cloud & DevOps** AWS (S3, Lambda, Textract, DynamoDB, CDK), Docker, K3s, Linux, Git, CI/CD, REST APIs

## Education

### University of Waterloo

Waterloo, Ont.

Master's in Applied Mathematics - Thesis: H(div)-conforming DG methods for multiphase flow

2021

### Thompson Rivers University

Kamloops, B.C.

B.S. in Computer Science and Mathematics - GPA: 4.21 / 4.33

2018

## Publications

**Edge-coloured graph homomorphisms, alternating paths and duality** — JCMCC

**On accuracy of numerical solutions to BVPs on infinite domains with slow decay** — MMNP

## Projects

### AI Movie Recommender (Serverless, AWS)

Vancouver, B.C.

Personal Project

May 2026

- Built a serverless AI movie recommender on AWS by ingesting watch history into DynamoDB and generates personalized options with Amazon Bedrock (Nova Micro) behind Cognito-authenticated API Gateway and Lambda endpoints.

### Self-Hosted Server

Vancouver, B.C.

Home-Lab Infrastructure

2023 - Present

- Built and operate a Linux home lab running 100+ Docker containers on K3s and Portainer, serving 50+ active users at 99% uptime.
- Self-hosted PostgreSQL and Gitea backing internal services; automated backups, container updates, and incident recovery.